Srajan Goyal

PhD Researcher · University of Trento and Fondazione Bruno Kessler
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Research Interests

My work combines symbolic and probabilistic verification, assume–guarantee contracts and closed-loop simulation, with applications to autonomous driving and space robotics. My interests include formal methods for AI safety, specification-guided evaluation, behavioural monitoring and adversarial testing for autonomous systems.

Education

PhD in Computer Science **Nov 2022 – present**
University of Trento, Italy; research conducted at Fondazione Bruno Kessler

Thesis: Simulation-based Verification and Validation of AI-enhanced Autonomous Driving Systems
Advisor: Dr. Stefano Tonetta · Expected completion: January 2027

Visiting PhD Researcher **Mar 2025 – Jun 2025**
University of York, United Kingdom

Advisor: Prof. Radu Calinescu

MSc in Automotive Technology **2014 – 2016**
Eindhoven University of Technology, Netherlands

Specialisation: Control Systems Technology

BEng in Mechanical Engineering **2008 – 2012**
VIT University, Vellore, India

Publications

Formal Methods and Autonomous Systems

Journal Articles

2025 **S. Goyal**, A. Griggio and S. Tonetta.
System-level Simulation-based Verification of Autonomous Driving Systems with the VIVAS Framework and CARLA Simulator.
Science of Computer Programming, 242, 103253. DOI

Conference Papers

2026 **S. Goyal**, R. Calinescu, A. Griggio, I. Stefanakos and S. Tonetta.
Policy-Guided Test Generation for Autonomous Driving Systems via Probabilistic Model Checking.
Software Engineering and Formal Methods (SEFM 2026). Accepted. Paper

2024 **S. Goyal**, A. Griggio and S. Tonetta.
Leveraging Contracts for Failure Monitoring and Identification in Automated Driving Systems.
Software Engineering and Formal Methods (SEFM 2024), LNCS 15280, pp. 441–460. DOI

2023 S. Fratini, P. Fleith, N. Policella, A. Griggio, S. Tonetta, **S. Goyal**, et al.
Verification and Validation of Autonomous Systems with Embedded AI: The VIVAS Approach.
17th Symposium on Advanced Space Technologies in Automation and Robotics (ASTRA), ESA-ESTEC.

2023 M. Bozzano, R. Bussola, M. Cristoforetti, **S. Goyal**, M. Jonáš, K. Kapellos, A. Micheli, D. Soldà, S. Tonetta, C. Tranoris and A. Valentini.
RobDT: AI-enhanced Digital Twin for Space Exploration Robotic Assets.
Applied Intelligence and Informatics (AII 2022), The Use of Artificial Intelligence for Space Applications, Studies in Computational Intelligence 1088, pp. 183–198. DOI

Workshop Papers (Peer-reviewed Proceedings)

2023 **S. Goyal**, A. Griggio, J. Kimblad and S. Tonetta.
Automatic Generation of Scenarios for System-level Simulation-based Verification of Autonomous Driving Systems.
Formal Methods for Autonomous Systems (FMAS 2023), EPTCS 395, pp. 113–129. DOI

Earlier Work in Control and Modelling

Conference Papers

2019 W. De Groote, E. Kikken, **S. Goyal**, S. Van Hoecke, E. Hostens and G. Crevecoeur.
Hybrid Derivative Functions for Identification of Unknown Loads and Physical Parameters with Application on Slider-Crank Mechanism.
IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM), pp. 1049–1054.

2018 **S. Goyal**, F. Verbelen, A. A. Abdallh, K. Stockman and P. Sergeant.
Energy Management Strategy for Oscillating Drivetrains Equipped with an Electric Variable Transmission.
International Conference on Electrical Machines and Systems (ICEMS), pp. 2145–2150.

2018 L. Mine, **S. Goyal** and A. A. E. Abdallh.
Improved Dynamic Modeling of Squirrel-Cage Induction Machines Using Magnetic Reluctance Network Theory.
International Conference on Electrical Machines and Systems (ICEMS), pp. 555–559. DOI

Book Chapters

2018 E. Feru, **S. Goyal** and F. Willems.
Optimal Sizing of Waste Heat Recovery Systems for Dynamic Engine Conditions.
Book chapter in Organic Rankine Cycle Technology for Heat Recovery. DOI

Earlier Research and Engineering Experience

Research Engineer / Software Developer **May 2021 – Oct 2022**

Fondazione Bruno Kessler, Trento, Italy

European Space Agency projects: RobDT and VIVAS on ExoMars rover autonomy

- Led the state-of-the-art technical survey and requirements analysis for an AI-enhanced digital twin of an ExoMars rover, synthesising research on machine learning, planning and scheduling, what-if analysis, anomaly detection and formal verification into the project's technical baseline.
- Co-developed the generic VIVAS methodology for simulation-based verification and validation of ExoMars rover autonomy, including symbolic behavioural models for resource prediction and opportunistic science applications.

Associate Research Engineer, Decisions and Control **Apr 2017 – Aug 2020**

Flanders Make, Lommel, Belgium

- Developed and validated online motion-planning and control algorithms for Level-4 autonomous mobile systems operating around stochastic moving obstacles and human workers.
- Deployed collision-avoidance and steering algorithms on a physical L4 autonomous guided vehicle. Designed edge-case experiments that exposed failures in perception, system integration and collision avoidance beyond nominal tests.
- Developed hybrid physics-informed machine-learning models for industrial digital twins, unknown-load identification and adaptive manufacturing processes.

MSc Researcher, Powertrains **Sep 2015 – Dec 2016**

TNO Automotive, Helmond, Netherlands

- Developed scalable dynamic models and joint design–control optimisation for an organic Rankine cycle waste-heat-recovery system.
- Developed switching model-predictive control that reduced simulated Euro-VI heavy-duty powertrain fuel consumption by 1.1%.

Engineer, Research and Development, Powertrain **May 2013 – Jul 2014**

Denso International India, Gurugram, India

- Calibrated and validated production and prototype automotive control systems for fuel-system monitoring, emissions compliance and bi-fuel operation.

Scientific Activities

Conference and Workshop Presentations

2024 Leveraging Contracts for Failure Monitoring and Identification in Automated Driving Systems.
Paper presentation, Software Engineering and Formal Methods (SEFM 2024).

2023 Automatic Generation of Scenarios for System-level Simulation-based Verification of Autonomous Driving Systems.
Paper presentation, Formal Methods for Autonomous Systems (FMAS 2023).

Poster Presentations

2025 Verification & Validation of AI-Based Autonomous Systems.
International Workshop on Digital Twins: Mathematical Analysis, Formal Methods and Scientific Machine Learning, University of Padova, Italy · February 2025.

2024 Probability in Computer Science (PICS) PhD School.
IT University of Copenhagen, Denmark · October 2024.

Peer Review

Conference reviewer. Forum on Specification and Design Languages (FDL 2026); International Symposium on Formal Methods (FM 2024); NASA Formal Methods Symposium (NFM 2023).

Journal reviewer. Science of Computer Programming.